

Yu-Sen Wu

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EDUCATION

Northwestern University

Master of Science in Machine Learning and Data Science

Evanston, IL, USA

Sep 2025 – Expected Dec 2026

National Tsing Hua University (NTHU)

Bachelor & Master of Science in Industrial Engineering & Engineering Management, GPA(4.0/4.3)

Hsinchu, Taiwan

Sep 2019 – Jun 2024

Technical Skills

- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, AI agent, LangChain, RAG
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI, Jupyter Notebook, Excel
- **MLOps & Cloud:** MLflow, Kubeflow, SageMaker, AWS, Docker, Git, CI/CD Pipelines
- **Programming:** Python, SQL, R, JavaScript, C#

Work Experience

Data Analyst Intern

WOOX - Global Crypto Exchange

Jun 2025 – Present

- Built automated text analytics pipeline to process regulatory documents from 19 crypto exchanges, using text similarity algorithms that reduced manual compliance analysis time by 95%
- Developed RAG-powered conversational AI system for natural language database querying with automated validation, reducing response time from hours to seconds while maintaining 97%+ accuracy
- Created Tableau dashboards to visualize user verification metrics, conducting funnel analysis and cohort analysis to identify operational bottlenecks and trends

Machine Learning Researcher

Ergo Sensing Lab

Jul 2022 – Dec 2024

- Built a multimodal ML framework combining computer vision and time-series data for user identification with dimensionality reduction and feature fusion, achieving 99% accuracy in real-time inference
- Designed controlled statistical experiments implementing PCA, signal filtering, and feature extraction, improving baseline model performance by 31%
- Developed real-time ML pipeline using time-series analysis and anomaly detection for fall prediction, achieving 96% accuracy across 200+ deployed industrial workers

Selected Projects

AI-Powered Educational Analytics Platform

Feb 2025 – Sep 2025

- Built recommendation engine using collaborative filtering and content-based algorithms with similarity metrics to solve cold-start problems, achieving 40% improvement in student-tutor pairing success rates
- Eliminated guesswork in UI/UX decisions by constructing A/B testing, driving 25% user engagement increase through data-driven interface optimizations

Time-Series Anomaly Detection for UV Oven Lamp Failure Prediction

Sep 2023 – Nov 2023

- Selected as finalist (top 10%) from 153 competing teams in *National Intelligent Manufacturing & Big Data Analytics Contest*
- Implemented PCA, correlation analysis, and t-tests on multivariate sensor data (temperature, voltage, current, vibration) to identify failure predictors with 95% statistical significance
- Built ensemble model combining LSTM and Isolation Forest with cross-validation optimization, achieving 96% precision and 91% recall for 72-hour advance predictions

Publication

Utilization of Two Types of Feature Datasets with Image-based and Time Series Deep Learning Models in Recognizing Walking Status and Revealing Personal Identification

Oct 2024

Yun-Ju Lee, **Yu-Sen Wu**, Pu-Chun Lin

Advanced Engineering Informatics, Vol. 62, Part B, Article 102729 (Impact Factor: 9.9, Top 10% in Computer Science)

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